THE UNIVERSITY OF HONG KONG
DEPARTMENT OF MATHEMATICS
MATH3603 Probability Theory

Tutorial 4
2026/09/29

1. Show that if f and g are density functions and $0 \leq \lambda \leq 1$, then $\lambda f + (1-\lambda)g$ is also a density function. Is the product fg still a density function?

2. Find the expected value and variance of a random variable X when

   (a) $X \sim Geom(p)$;

   (b) $X \sim NB(r, p)$;

   (c) $X \sim Poisson(\lambda)$;

   (d) $X \sim U[a, b]$;

   (e) $X \sim Exp(\lambda)$;

   (f) $X \sim Gamma(\alpha, \lambda)$;

3. Prove the Inverse transform sampling for general random variable X. i.e. For any random variable X (with c.d.f F) and $U \sim U[0, 1]$, the random variable $F^{-1}(U)$ has the same distribution as X, where $F^{-1} : [0, 1] \to \mathbb{R}$ is defined as

$$F^{-1}(u) = \inf\{x \mid F(x) \geq u\}.$$